

MCCLELLANVILLE 7 NE, SC, USA

WMO: 747850

Lat: 33.1532N

Lon: 79.3637W

Elev: 9

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 03728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 27.0	(c) 30.3	(d) 11.9	(e) 10.1	(f) 31.5	(g) 17.0	(h) 13.0	(i) 35.0	(j) 6.8	(k) 59.4	(l) 6.5	(m) 58.6	(n) 1.4	(o) N/A	(p) 0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.8	(c) 91.1	(d) 78.3	(e) 89.1	(f) 77.8	(g) 87.4	(h) 77.0	(i) 80.3	(j) 87.8	(k) 79.4	(l) 86.5	(m) 78.5	(n) 85.2	(o) 3.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.3	146.8	84.5	77.4	142.5	83.4	76.7	139.2	82.8	43.7	88.0	42.8	86.7	42.0	85.3	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.4	(b) 5.4	(c) 4.8	(e) 22.0	(f) 95.6	(g) 3.9	(h) 2.2	(i) 19.3	(j) 97.2	(k) 17.0	(l) 98.5	(m) 14.8	(n) 99.7	(o) 12.0	(p) 101.4		
(a) 19.6	(b) 82.4	(c) 4.0	(e) 19.6	(f) 82.4	(g) 4.0	(h) 0.9	(i) 16.7	(j) 83.0	(k) 14.4	(l) 83.5	(m) 12.2	(n) 84.0	(o) 9.3	(p) 84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.3	48.1	50.7	56.7	64.2	71.8	78.2	80.3	79.9	76.0	67.2	56.9	52.1
	DBStd	13.34	9.78	8.44	8.26	6.64	5.26	3.60	2.80	2.97	3.99	7.01	7.95	8.83
	HDD50	390	159	88	34	2	0	0	0	0	0	1	25	81
	HDD65	2039	526	403	274	92	11	0	0	0	1	61	263	408
	CDD50	5959	99	108	242	429	677	847	939	927	780	534	231	146
	CDD65	2134	2	3	17	69	224	397	474	462	330	129	19	8
	CDH74	16957	3	8	62	278	1314	3481	4594	4197	2308	653	50	9
	CDH80	4990	0	0	5	28	249	1070	1591	1395	569	82	1	0
(14) Wind	WSAvg	2.1	2.0	2.2	2.5	2.6	2.4	2.2	2.0	1.7	1.8	1.7	1.7	1.8
(15) Precipitation	PrecAvg	57.7	4.3	3.2	4.5	3.1	3.3	6.0	7.0	7.5	6.4	5.4	2.8	3.7
	PrecMax	69.4	10.1	11.4	11.0	8.0	6.8	9.9	10.9	17.0	16.5	11.7	7.9	11.3
	PrecMin	41.0	0.6	1.1	1.1	0.6	1.8	2.5	2.8	2.1	0.9	0.0	0.4	1.1
	PrecStd	8.4	2.7	2.3	2.4	2.0	1.3	2.5	2.4	3.9	4.4	3.5	2.0	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.2	75.2	79.3	83.4	90.0	93.2	93.4	93.1	89.0	84.9	78.7	75.3
		MCWB	65.2	66.0	62.6	67.7	73.3	78.2	79.2	79.2	75.9	74.6	70.1	68.7
	2%	DB	69.4	70.9	75.5	79.4	85.6	90.1	90.9	90.2	86.8	82.3	75.1	71.8
		MCWB	63.1	62.4	64.7	68.9	72.8	77.7	78.8	78.6	75.8	73.1	67.2	66.9
	5%	DB	65.7	67.2	72.4	77.2	82.4	87.6	88.9	88.3	85.4	80.1	72.5	68.9
		MCWB	60.8	61.0	62.2	67.5	72.2	77.0	78.4	77.8	75.2	71.6	66.1	64.7
	10%	DB	62.8	64.1	69.1	75.0	80.3	85.7	87.1	86.6	83.7	77.8	69.8	66.0
		MCWB	58.5	59.1	61.2	66.2	71.4	75.9	77.6	77.1	74.4	70.5	64.5	62.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	67.9	68.4	73.6	77.1	80.6	81.9	81.6	79.1	77.3	71.7	71.0
		MCDB	71.1	74.3	74.9	78.1	85.5	89.4	90.1	88.9	84.8	81.6	75.3	74.0
	2%	WB	64.7	65.4	66.6	71.6	75.1	79.4	80.4	80.2	78.0	76.0	69.3	68.7
		MCDB	68.4	69.9	72.9	76.9	81.5	88.0	88.0	87.0	83.7	80.5	72.3	71.1
	5%	WB	62.3	63.5	64.8	70.1	73.9	78.3	79.5	79.2	77.1	74.4	67.7	66.0
		MCDB	65.2	67.3	70.6	75.5	80.3	85.9	86.7	85.6	82.8	78.8	71.0	68.8
	10%	WB	59.3	61.0	63.1	68.4	72.8	77.4	78.5	78.3	76.1	72.5	65.4	63.4
		MCDB	61.7	64.6	68.0	73.5	78.9	84.2	85.3	84.5	81.7	77.0	69.4	66.3
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	17.6	17.7	17.3	15.2	14.1	13.8	13.0	13.7	16.2	18.3	17.1
		MCDBR	18.5	19.9	19.8	17.5	16.6	16.3	15.4	14.8	14.9	16.1	18.3	16.8
		MCWBR	13.3	12.4	10.6	8.8	7.4	7.2	7.0	6.3	6.5	8.4	11.3	12.5
	5% WB	MCDBR	16.6	17.6	16.7	14.2	13.9	15.1	14.5	13.6	12.9	13.3	15.4	15.4
		MCWBR	14.0	13.4	9.8	8.6	6.8	7.3	7.2	6.2	6.2	8.3	11.5	12.9
(40) Clear-Sky Solar Irradiance	taub		0.331	0.338	0.367	0.403	0.469	0.488	0.532	0.508	0.434	0.393	0.351	0.333
	taud		2.527	2.507	2.447	2.323	2.195	2.191	2.080	2.143	2.345	2.436	2.508	2.551
	Ebn at Noon		281	291	289	281	263	257	245	248	264	267	271	274
	Edh at Noon		27	30	34	41	47	47	52	48	38	32	27	25
(44) All-Sky Solar Radiation	RadAvg		883	1094	1464	1878	2036	2075	1978	1777	1515	1290	985	789
	RadStd		55	113	134	130	140	143	120	103	124	138	72	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.46	N/A	N/A	N/A	N/A	N/A	N/A	+461	+288
(47) Regional (41 neighbors)		+1.03	N/A	N/A	N/A	N/A	N/A	N/A	+349	+247

Nomenclature: See separate page